

Phase 7 — Nursing Station + Supervisor Telemetry/ECG/Intercom (M22–M29)

Status: ■ **PHASE 7 COMPLETE** (2026-05-26) **Combined guide** — eight modules + five 1.x touch-ups shipped in one autonomous pass. Each section below maps to one module from the Phase 7 plan.

Pre-Phase 7 touch-ups (1.1–1.5)

#	What	Files
1.1	"Charge-nurse dashboard" → "Multi-Patient Control" rename (label-only)	templates/control_room.html, templates/base.html
1.2	Student.role field + schema migration v5 (student.role default 'bedside') + students_by_role helper	ehr_db.py, control_room.py
1.3	M5 dashboard card click now routes to /portal/room/encounter/{id} (M22 target)	static/control_room.js
1.4	M7 alarm-class scenes (code.blue, pump.alarm) tag payloads with level='alarm' so the M26 bus picks them up	scenes.py
1.5	devices.registry.list_kinds() helper	devices/registry.py

Two M1 tests updated to be future-proof (schema_version >= 4 instead of == 4).

M22 — Per-Patient Console scaffold

Goal: instructor drill-in at /portal/room/encounter/{id} — new template + JS + CSS scaffold with 6 cards (telemetry, ECG, device list, telemetry overrides, scene injector, recent chart events). M22 ships the scaffold; M23/M24/M25 light up each card.

Files: templates/encounter_console.html, static/encounter_console.{js,css}, server.py route. **4 tests pass** (route renders, 404s on unknown id, 404s when no active room, clone indicator surfaces).

M23 — Telemetry simulation engine

Goal: derive a continuous HR/BP/SpO2/RR/Temp snapshot per encounter from the latest vitals.record event + small jitter, with operator force-set overrides per metric.

Files: portal/telemetry.py (new), routes added to server.py (GET /api/encounter/{id}/telemetry + POST /api/encounter/{id}/telemetry/override). Overrides live in-memory on the Encounter.telemetry_overrides dict (the room dies with the server; no restart-survival needed). **5 tests pass** (snapshot from latest vitals, defaults fallback, scene injection shifts snapshot, overrides take precedence, route round-trip set/clear/clear_all).

M24 — ECG waveform library

Goal: static catalog of 11 cardiac rhythms (NSR, sinus tachy/ brady, AFib, AFlutter, monomorphic VT, polymorphic VT/torsades, VFib, asystole, PEA, paced). Per-encounter `ecg_rhythm_id` + `ecg_enabled` toggle. Client-side SVG renderer scrolls a 6-second strip.

Files: `portal/ecg.py` (new) + `static/ecg_strip.js` (new client renderer). Routes: GET `/api/ecg/catalog`, GET `/api/encounter/{id}/ecg`, POST `/api/encounter/{id}/ecg`. **8 tests pass** (catalog has 11 with required fields, get/lookup helpers, encounter defaults, route round-trip, unknown-id rejected, observer can't mutate).

M25 — Per-Patient Console rich features

Goal: light up M22's scaffold with live telemetry poll (1s), ECG strip via `ecg_strip.js` + picker dropdown, device list from `/api/room/state`, telemetry override sliders (one per metric), override-aware coloring. Scene injector preserved from M22.

Files: rewritten `static/encounter_console.js` + template includes `ecg_strip.js`. **2 tests pass** (template references renderer + the right anchors; end-to-end telemetry route round-trip).

M26 — Alarm bus

Goal: unified active-alarm list aggregating three sources — device-event `alarm.injected` (v6 pump/cabinet), chart-event `level='alarm'` payloads (M7 + Phase 7 1.4), and future-device button presses (M29). Each alarm carries `severity` (info/warning/critical), `source`, `kind`, `encounter_id`, `alarm_id` (deterministic, stable across reads).

Files: `portal/alarms.py` (new), routes GET `/api/room/alarms` + POST `/api/alarm/{id}/clear`. **6 tests pass** (pump device alarm surfaces, code.blue scene alarm surfaces, clear removes from list, critical-first ordering, unknown id 404, no-room 404).

M27 — Nursing Station student role

Goal: new in-sim role — supervisor monitors every bed remotely. Student-join page (M9) gains a role-picker step (Bedside vs Nursing Station). `Nurse_station` register skips encounter assignment + chat station creation; the student lands on `/portal/students/nurse_station?sid=...` with one bed card per encounter (mini telemetry strip + mini ECG + device pills + alarm board at top).

Files: new `templates/nurse_station.html`, `static/nurse_station.{js,css}`, role-picker step in `templates/student_join.html`, role branching in `static/student_join.js`, new POST `/portal/students/register_nurse` + GET `/portal/students/nurse_station` routes. **7 tests pass**.

M28 — Intercom (one-way nurse → bedside)

Goal: nurse-station types a message → server writes a `comm.intercom` chart_event with `text` + `voice_id` + `persona_id` (staff persona auto-picked from encounter's `selected_personas` when available); WS push (M16) on `/ws/room/{room_code}` so the bedside chat station can play/show it. Bedside replies via existing v6 chat path (text).

Files: `portal/intercom.py` (new), POST `/api/intercom/{encounter_id}/page` route. Future-WebRTC upgrade preserves the data contract. **7 tests pass** (comm event recorded, staff voice when

bound, no-voice fallback, route auth — 403 if from_student_id is bedside, 400 on empty text, 404 on unknown encounter, WS push fires).

M29 — Future-device stubs

Goal: four new in-sim device kinds — `call_bell`, `bed_alarm`, `code_blue_button`, `fire_alarm`. Each is a single-button bedside press that emits an alarm. injected device_event with `tone=kind`. M26's severity table classifies them appropriately (`call_bell` → info, `bed_alarm` → warning, `code_blue_button` + `fire_alarm` → critical).

Files: `portal/future_devices.py` (new), routes `GET /api/future_devices/kinds` + `POST /api/encounter/{id}/future_device/{kind}/press`. **9 tests pass** (four kinds listed, each kind raises the right severity, unknown kind 400, unknown encounter 404, no-room 404, WS push fires).

Combined test summary

Module	New v7 tests
1.1–1.5 touch-ups (no new tests; updated 2 M1 brittle assertions)	0
M22	4
M23	5
M24	8
M25	2
M26	6
M27	7
M28	7
M29	9
Total Phase 7	48

v7 suite at start of Phase 7: 132. v7 suite after Phase 7: 180 passed, 1 Playwright skip. Full v6 regression on v7: 291 passed, 6 env-flaky, 2 skipped, 0 v7 regressions (matches v6 baseline).

Architecture overview after Phase 7

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INSTRUCTOR (Operator) – /portal/room and children
■■■ /portal/control      (wizard – Single Patient / Room of N)
■■■ /portal/room        (Multi-Patient Control – was "Charge-nurse")
■   ■■■ encounter grid + Freeze/Resume/Scene/End controls
■   ■■■ card-click → /portal/room/encounter/{id}
■■■ /portal/room/encounter/{id} (Per-Patient Console – NEW M22+M25)
    ■■■ live telemetry strip      (M23)
    ■■■ ECG strip + picker        (M24)
    ■■■ device list               (M1.5 + M25)
    ■■■ telemetry overrides       (M23)
    ■■■ pre-targeted scene inject (M7)
    ■■■ recent chart events

STUDENT (in-simulation roles)
■■■ Bedside (existing M9)
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- ■■■ chat station (v3)
- ■■■ EHR station (v5)
- ■■■ device station (v6 pumps/cabinets)
- + Phase 7 M29 stubs:
- call_bell, bed_alarm,
- code_blue_button, fire_alarm
- /portal/students/nurse_station (NEW M27)
- multi-patient telemetry mini-strips (M23)
- mini ECG per bed (M24)
- device pills per bed
- alarm board (M26)
- intercom → any bed (M28)

What still needs the operator

1. **LAN_TEST_V7.md sign-off** (still M21's open item) — now also covers Phase 7 surfaces. 2. **JS subscriber wiring on bedside chat station** for M28 intercom audio playback. The server-side contract + WS push are live; the bedside JS hook (~15 lines in `station_chat.js`) is the only remaining wiring. 3. **v7.1 WebRTC upgrade for M28 intercom** — two-way mic-to-mic. Data contract is forward-compatible.

Combined change list

Date	Module	Files
2026-05-26	1.1 rename	templates/control_room.html, templates/base.html
2026-05-26	1.2 Student.role + schema v5	ehr_db.py, control_room.py
2026-05-26	1.3 drill-in route swap	static/control_room.js
2026-05-26	1.4 scene alarm level tag	scenes.py
2026-05-26	1.5 device kinds API	devices/registry.py
2026-05-26	M22	templates/encounter_console.html, static/encounter_console.{js,css}, server.py route
2026-05-26	M23	portal/telemetry.py (new), server.py routes, control_session.py field
2026-05-26	M24	portal/ecg.py (new), static/ecg_strip.js (new), server.py routes, control_session.py fields
2026-05-26	M25	rewritten static/encounter_console.js, template includes
2026-05-26	M26	portal/alarms.py (new), server.py routes
2026-05-26	M27	templates/nurse_station.html, static/nurse_station.{js,css}, role step in student_join.html, role branch in student_join.js, server.py routes
2026-05-26	M28	portal/intercom.py (new), server.py route + WS push

Date	Module	Files
2026-05-26	M29	portal/future_devices.py (new), server.py routes + WS push

Open follow-ups

- Bedside chat station JS intercom playback hook (~15 lines).
- WebRTC v7.1 upgrade for two-way intercom audio.
- LAN_TEST_V7.md operator sign-off (now covers Phase 7 surfaces).
- The two voices test fixtures + four device_debrief tests (the pre-existing v6 env-flaky 6) — independent hygiene pass.